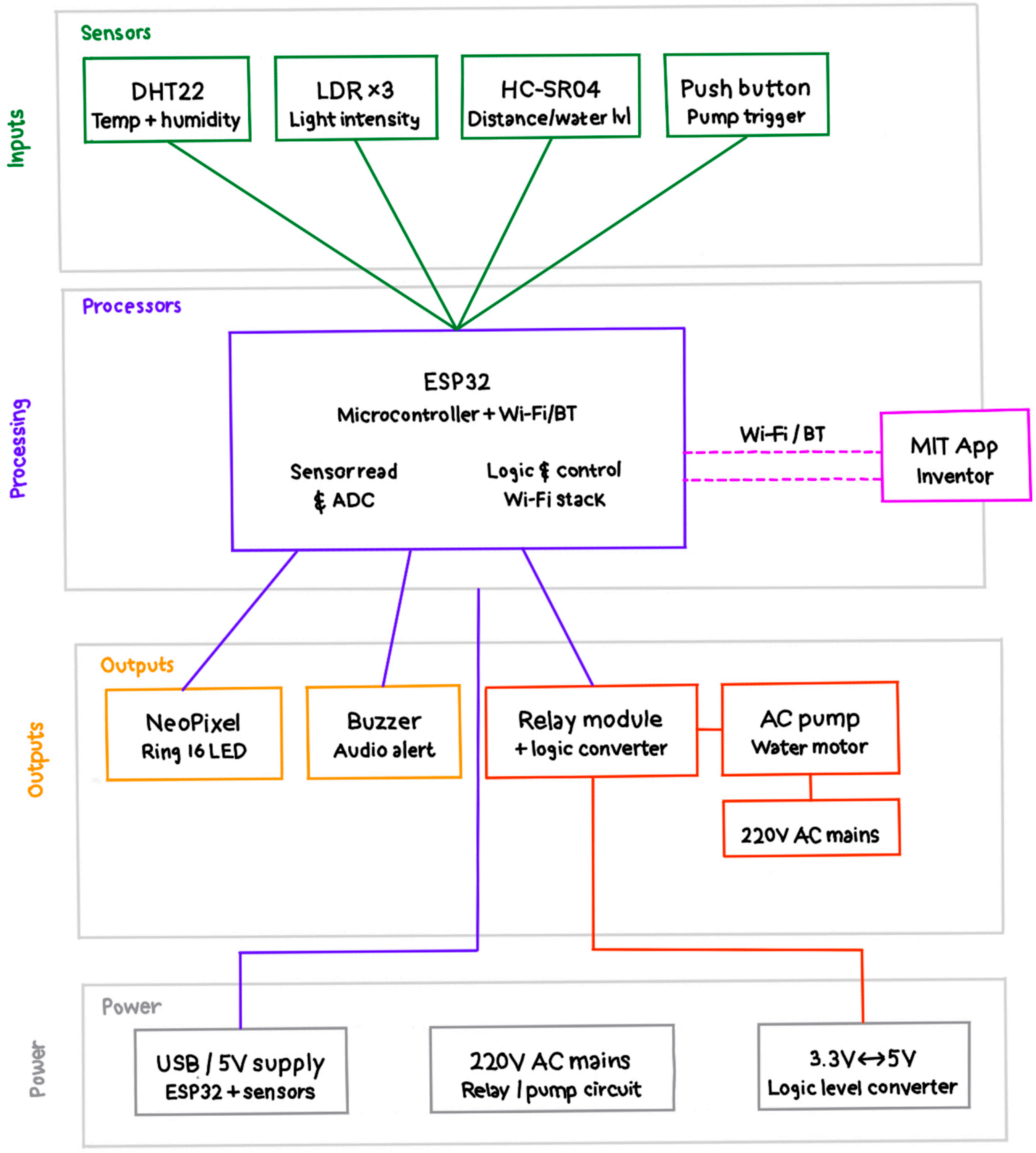
Inputs
Sensors
DHT22
Temp + humidity
LDR ×3
Light intensity
HC-SR04
Distance/water lvl
Push button
Pump trigger
Processing
Processors
ESP32
Microcontroller + Wi-Fi/BT
Sensor read
& ADC
Logic & control
Wi-Fi stack
Wi-Fi / BT
MIT App
Inventor
Outputs
Outputs
NeoPixel
Ring 16 LED
Buzzer
Audio alert
Relay module
+ logic converter
AC pump
Water motor
220V AC mains
Power
Power
USB / 5V supply
ESP32 + sensors
220V AC mains
Relay / pump circuit
3.3V↔5V
Logic level converter